

Assignment - 03

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Q1. Solution :- When three coins are tossed, the total number of outcomes is $2^3 = 8$.

Sample Space (S) : $\{ HHH, HHT, HTH, THH, HTT, THT, TTH, TTT \}$

Now, let's consider the event of getting at least two heads. Outcomes :-

$E \Rightarrow \{ HHH, HHT, HTH, THH \}$

So, Elements in the sample space (S) $\Rightarrow 8$
" " " event (E) $\Rightarrow 4$

Q2. Solution :- When two dice are thrown simultaneously, the total number of outcomes $(6 \times 6) = 36$.

To find the probability of obtaining a total score less than 11, we need to determine the favorable outcomes.

All the possible outcomes :

(1,1), (1,2) - - - (1,6)
⋮
(6,1), (6,2) - - - (6,6)

Now, let's identify the outcomes where the total score is less than 11:

$(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)$
 $(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)$
 $(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)$
 $(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)$
 $(5,1), (5,2), (5,3), (5,4), (5,5), (6,1)$
 $(6,2), (6,3), (6,4)$

Total outcomes where the total score is less than 11 $\Rightarrow 33$

Probability $= \frac{33}{36} = \frac{11}{12}$

Q.3. Solution + In a leap year, total days $= 366$

To determine the chance that it will contain 53 Sundays, we need to consider the number of days in a week, which is 7. The number of days in a year is not evenly divisible by the number of days in a week, so there will be an extra day.

For 53 Sundays to occur in a leap year, there must be one day where there are two Sundays. This can happen in one of the following ways:-

1. The extra day is a Sunday.
2. The year starts on a Sunday.

Let's calculate the probability for each case:

(i) Probability that the extra day is a Sunday: $\frac{1}{7}$

(ii) Probability that the year starts on a Sunday: $\frac{1}{7}$

Total probability $= \frac{1}{7} + \frac{1}{7}$

$\Rightarrow \frac{2}{7}$

$\Rightarrow$ Therefore, the chance that a randomly selected leap year will contain 53 Sundays is $\frac{2}{7}$

Q.4. Solution:

(a). Probability that all three balls are white:

$P(W, W, W) = \frac{\text{Number of ways to choose 3 white balls}}{\text{Total number of ways to choose 3 balls}}$

$$\Rightarrow \frac{5/3}{43/3}$$

$$P(W, W, W) \Rightarrow \frac{5}{43}$$

(b). Probability that all three balls are red :

$$P(R,R,R) = \frac{\text{Number of ways to choose 3 red balls}}{\text{Total number of ways to choose 3 balls}}$$

$$P(R,R,R) \Rightarrow \frac{8/3}{13/3}$$

$$P(R,R,R) \Rightarrow \frac{8}{13} \quad \Delta$$

(c). Probability that one ball is red and two balls are white :

$$P(R,W,W) + P(W,R,W) + P(W,W,R)$$

$$P(R,W,W) = \frac{\binom{8}{1} \cdot \binom{5}{2}}{\binom{13}{3}} = \frac{20 \times 3}{13} = \frac{60}{13}$$

$$P(W,R,W) = \frac{\binom{8}{1} \cdot \binom{5}{2}}{\binom{13}{3}} = \frac{60}{13}$$

$$P(W,W,R) = \frac{\binom{8}{1} \cdot \binom{5}{2}}{\binom{13}{3}} = \frac{60}{13}$$

$$\Rightarrow \frac{60}{13} + \frac{60}{13} + \frac{60}{13}$$

$$P(\text{One red \& two white}) \Rightarrow \frac{180}{13} \quad \Delta$$

Solution :- Let,

- A : Passing test I
- B : Passing test II
- C : Passing test III

The given information can be expressed as probability

$$P(A) = p$$

$$P(B) = q$$

$$P(C) = \frac{1}{2}$$

The student is successful if he passes either tests I and II or tests I and III.

$$P(\text{Success}) \Rightarrow P[(A \cap B) \cup (A \cap C)]$$

Using the probability addition rule, we have,

$$P(\text{Success}) \Rightarrow P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$$

$$P(\text{Success}) \Rightarrow P(A \cap B) + P(A \cap C) - P(A \cap B \cap C)$$

Now, we can substitute the given probabilities:

$$P(\text{Success}) \Rightarrow p \cdot q + p \cdot \frac{1}{2} - p \cdot q$$

Combine like terms:

$$P(\text{Success}) = p \cdot \frac{1}{2}$$

Now, it's given that the probability of success is $\frac{1}{2}$.

$$\frac{1}{2} = p \cdot \frac{1}{2}$$

Solving for p .

$$p \Rightarrow 1$$

So, the relation between p and q is $p = 1 - q$.

$$p = 1 - q$$

Q6 Solution :- Let's denote the events as follows.

A = Occurrence of event A

B = Occurrence of event B

The given information can be expressed as probability.

$$P(A \cap B) \Rightarrow \frac{1}{12}$$

$$P(A' \cap B') \Rightarrow \frac{1}{12}$$

Complement of Event A = A'

Complement of Event B = B'

We know that, $P(A \cap B) = P(A) \cdot P(B)$ ($\because A$ & B are independent)

$$P(A \cap B) = P(A) \cdot P(B) = \frac{1}{12}$$

Probability for A occurs and B does not occur:

$$P(A \cap B') = P(A) \cdot P(B')$$

$$\Rightarrow P(A) \cdot (1 - P(B))$$

Substitute the values:

$$P(A \cap B') = \frac{1}{12} \cdot (1 - P(B))$$

Similarly, when B occurs and A does not

$$P(A' \cap B) = (1 - P(A)) \cdot P(B)$$

Substitute the values:

$$P(A' \cap B) = (1 - \frac{1}{12}) \cdot \frac{1}{12}$$

Now, we know that the probability of neither A nor B happening is $P(A' \cap B')$:

$$P(A' \cap B') = \frac{1}{12}$$

We know:-

$$P(A' \cap B') = P(A') \cdot P(B')$$

$$\frac{1}{12} = [1 - P(A)] \cdot [1 - P(B)]$$

$$\left(\frac{1}{2} \cdot \frac{1}{2}\right) \cdot \left(\frac{1}{2}\right) = \frac{1}{8}$$

Now, we can solve for $P(A \cap B)$

$$P(A \cap B) \Rightarrow \frac{1}{8}$$

So, the probability of the occurrence of both A and B is $\frac{1}{8}$.

Q7. Solution: Let's denote the probabilities as follows.

$$\begin{aligned} P(A \cap B \cap C) &= 1 - a \\ P(\bar{A} \cap B \cap C) &= 1 - 2a \\ P(\bar{A} \cap \bar{B} \cap C) &= 1 - a \\ P(A \cap B \cap \bar{C}) &= a^2 \end{aligned}$$

Probability that at least one of the events A, B or C occurs $\Rightarrow P(A \cup B \cup C)$

Using the inclusion-exclusion principle, we have:

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$$

Now substitute:

$$\begin{aligned} P(A \cup B \cup C) &= P(A) + P(B) + P(C) - (1 - a) - (1 - a) - (1 - a) \\ &\quad - (1 - 2a) + a^2 \end{aligned}$$

$$\text{Since, } P(A) = P(B) = P(C)$$

(Symmetric)

$$\text{Let, } P(A) = P(B) = P(C) = x$$

$$P(A \cup B \cup C) = 3x - 2(1 - a) - (1 - 2a) + a^2$$

Simplify:

$$\begin{aligned} P(A \cup B \cup C) &= 3x - 2 + 2a - 1 + 2a - a^2 + a^2 \\ &\Rightarrow 4x - 3 + 4a \end{aligned}$$

Now, Since probabilities are between 0 and 1, x must be between 0 and $\frac{1}{3}$ to satisfy the given cond. Thus, the expression $4x - 3 + 4a$ will always be greater than $\frac{1}{2}$.

Therefore, Probability $\geq \frac{1}{2}$ Δ

Q8.

Solution: Let F be the event that the electronic component fails when first used, and L be the event that the component lasts for one year.

$$P(F) = 0.10$$

$$\begin{aligned} P(\text{Doesn't fail immediately}) &\Rightarrow 1 - P(F) \\ &= 1 - 0.10 \\ P(F') &\Rightarrow 0.90 \end{aligned}$$

$P(L|F') \Rightarrow$ (Probability that the component lasts for one year given that it does not fail immediately) $\Rightarrow 0.99$

Total probability $\div P(L)$

$$\Rightarrow P(F) \cdot P(L|F) + P(F') \cdot P(L|F')$$

$$P(L) \Rightarrow (0.10)(0) + (0.90)(0.99)$$

$$\Rightarrow 0 + 0.891$$

$$P(L) = 0.891 \text{ OR } 89.1\% \quad A$$

Q.9. Solution $\div$ Let's denote the events as follows:

- $W_A \div$ Group A wins the position on the board of directors
- $W_B \div$ Group B " " " " " "
- $W_C \div$ Group C " " " " " "
- $P \div$ A new product is introduced.

Probabilities $\div$

$$P(W_A) \Rightarrow 0.5$$

$$P(W_B) \Rightarrow 0.3$$

$$P(W_C) \Rightarrow 0.2$$

If group A wins, the probability of introducing a new product is

$$P(P|W_A) = 0.7$$

$$\text{for group B} \div P(P|W_B) = 0.6$$

$$\text{for group C} \div P(P|W_C) = 0.5$$

By laws of total probability $\div$

$$P(P) \Rightarrow P(W_A) \cdot P(P|W_A) + P(W_B) \cdot P(P|W_B) + P(W_C) \cdot P(P|W_C)$$

$$\Rightarrow (0.5 \times 0.7) + (0.3 \times 0.6) + (0.2 \times 0.5)$$

$$\Rightarrow 0.35 + 0.18 + 0.1$$

$$P(P) \Rightarrow 0.63 \text{ OR } 63\% \quad A$$

Q.10. Solution $\div$ Let's denote the events as follows:

- $W =$ Drawing a white ball
- $B =$ " " " black "

initially, the urn contains 2 white balls (W_1, W_2) and 2 black balls (B_1, B_2).

Q12. Solution - Let's denote the events as follows

- T : The man speaks the truth
 $\bar{T}$: The man doesn't speak the truth
 E : The man reports that the die shows a 6
 $\bar{E}$: " " " " " die doesn't show a 6

Given

$$P(T) = \frac{3}{4}$$

$$P(\bar{T}) = 1 - \frac{3}{4}$$

$$\Rightarrow \frac{1}{4}$$

Now,

$P(\text{Reporting a 6 given that the die actually shows a 6})$;

$$P(E|T) = \frac{1}{6}$$

$$\Rightarrow P(E|\bar{T}) = \frac{1}{6}$$

By Bayes's theorem -

$$P(T|E) \Rightarrow \frac{P(E|T) \cdot P(T)}{P(E)} \quad \text{--- (1)}$$

Q13.

Where,

$$P(E) = P(E|T) \cdot P(T) + P(E|\bar{T}) \cdot P(\bar{T})$$

$$P(E) \Rightarrow \frac{1}{6} \cdot \frac{3}{4} + \frac{1}{6} \cdot \frac{1}{4}$$

$$P(E) \Rightarrow \frac{4}{24}$$

Now put the value in eq. (1)

$$P(T|E) \Rightarrow \frac{\frac{1}{6} \cdot \frac{3}{4}}{\frac{4}{24}}$$

$$\Rightarrow \frac{1}{8} \cdot \frac{24}{4}$$

$$P(T|E) \Rightarrow \frac{3}{8}$$

Q13.

Solution - Let T be the event that the statement is true, and A and B be the events that witness A and B agree on the statement.

Given :-

$$P(T|A \cap B) \Rightarrow \frac{P(T \cap (A \cap B))}{P(A \cap B)}$$

Now, ~~that~~ we have.

$$P[T \cap (A \cap B)] = P(T) \cdot P(A \cap B)$$

Now, substitute it,

$$P(T/A \cap B) \Rightarrow \frac{P(T) \cdot P(A \cap B)}{P(A \cap B)} \text{--- (1)}$$

Now

$$\Rightarrow P(A \cap B) \Rightarrow P(A \cap B \cap T) + P(A \cap B \cap \bar{T})$$

$$P(A \cap B) \Rightarrow P(T) \cdot P(A) \cdot P(B) + P(\bar{T}) \cdot P(\bar{A}) \cdot P(B) \text{--- (2)}$$

Now, put the value of (1) in (2)

$$P(T/A \cap B) = \frac{P(T) \cdot P(A) \cdot P(B)}{P(T) \cdot P(A) \cdot P(B) + P(\bar{T}) \cdot P(\bar{A}) \cdot P(B)}$$

$$P(T/A \cap B) \Rightarrow 1$$

Q14. Solution :- Let's denote the events.

S: A ship reaching safely
D: A ship is drowning

Given, $P(D) \Rightarrow 1/10$

$$\text{So, } P(S) \Rightarrow 1 - P(D) \Rightarrow 9/10$$

Now we can break down the given condition in two cases:-

1. Exactly 4 ships reach safely.
2. All 5 ships reach safely.

Prob. for case 1:-

$$P(4S, 1D) \Rightarrow \left(\frac{5}{4}\right) \cdot P(S)^4 \cdot P(D) \Rightarrow 5 \cdot \left(\frac{9}{10}\right)^4 \cdot \frac{1}{10}$$

Prob for case 2:-

$$P(5S) \Rightarrow P(S)^5 \Rightarrow \left(\frac{9}{10}\right)^5$$

Now,

$$P(\text{At least 4 reach safely}) = P(4S, 1D) + P(5S)$$

$$P(\text{At least 4 reach safely}) = 5 \cdot \left(\frac{9}{10}\right)^4 \cdot \frac{1}{10} + \left(\frac{9}{10}\right)^5 \Rightarrow 5 \cdot \left(\frac{59049}{10000}\right) \cdot \frac{1}{10} + \left(\frac{59049}{10000}\right) \Rightarrow \frac{91854}{10000} \text{ OR } 0.91854$$

Q15.

Solution:- Let's denote the event of taking a step forward as F.
And the event of taking a step backward as B.

Given.

$$P(F) = 0.4$$

$$P(B) = 0.6$$

Let's calculate the probabilities for each case.

1. 6 steps forward, 5 steps Backward:

$$P(F, F, F, F, F, B, B, B, B, B) = (0.4)^6 \cdot (0.6)^5$$

2. 5 steps forward, 6 steps Backward:

$$P(F, F, F, F, B, B, B, B, B, B) = (0.4)^5 \cdot (0.6)^6$$

$\Rightarrow$ P (One step away after 11 steps) $\Rightarrow$

$$\Rightarrow (0.4)^6 \cdot (0.6)^5 + (0.4)^5 \cdot (0.6)^6$$

$$\Rightarrow 3.16768 \times 10^{-5} + 4.7616 \times 10^{-4}$$

$$\Rightarrow 0.0004782768$$

Solution:-

Possible outcomes $\div 2^3 = 8$ (3 coins)

Possible outcomes are:-

1. HHH 5. THH
2. HHT 6. THT
3. HTH 7. TTH
4. HTT 8. TTT

Let, X = Random variable representing the number of heads in a given outcome.

$$P(X=0) : P(\text{getting 0 heads}) = \frac{1}{8}$$

$$P(X=1) : \quad \quad \quad 1 \text{ heads} = \frac{3}{8}$$

$$P(X=2) : \quad \quad \quad 2 \text{ heads} = \frac{3}{8}$$

$$P(X=3) : \quad \quad \quad 3 \text{ heads} = \frac{1}{8}$$

In general

$$P(X=0) = \frac{1}{2^3}$$

$$P(X=1) = \frac{3}{2^3}$$

$$P(X=2) = \frac{3}{2^3}$$

$$P(X=3) = \frac{1}{2^3}$$

where 2^3 is the total number of possible outcomes when three coins are tossed

Q17.

Solution:- The mean (μ) and variance (σ^2) of a binomial distribution are related to the parameters n (number of trials) and p (prob. of success on a single trial) as follows:

$$\mu = np \quad \text{--- (i)}$$

$$\sigma^2 = np(1-p)$$

given,

$$\mu = 2$$

$$\sigma^2 = 1$$

$$\text{ii) } 2 = np$$

$$1 = np(1-p) \quad \text{--- (ii)}$$

from the 1st eq.

$$p = \frac{\mu}{n} \Rightarrow \frac{2}{n}$$

$$p = \frac{2}{n}$$

Put it in (ii) eq.

$$1 = \frac{2}{n} \cdot \left(1 - \frac{2}{n}\right)$$

$$n=2 \quad [n=2]$$

By using (i) eq.

$$p = \frac{\mu}{n} = \frac{2}{2}$$

$$\Rightarrow \frac{2}{2}$$

$$[p=1]$$

$P(X)$ takes values greater than 1 :-

$$P(X > 1) = P(X=2) + P(X=3) + \dots + [P(X=n)]$$

$$\text{for, } n=2 \quad \text{so } p=1$$

$$P(X=2) = 1$$

$$P(X=3) = 0$$

so,

$$P(X > 1) \Rightarrow 1+0$$

$$P(X > 1) \Rightarrow 1 \quad \text{✓}$$

Q18. Solution :- Let's denote,

$X =$ No. of defective bolts in a box

Given,

$p = 3\%$ of bolts are defective.
 $n = 50$ (No. of trials)

Prob. of single bolt being defective = 0.03

a). Exact Probability (Binomial Distribution):

PMF for Binomial Distribution.

$$P(X=k) \Rightarrow \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

for, $k=5$

$$P(X=5) \Rightarrow \binom{50}{5} \cdot (0.03)^5 \cdot (1-0.03)^{50-5}$$

$$P(X=5) \approx 0.218864$$

b). Poisson Approximation

The Poisson PMF is given by:

$$P(X=k) = \frac{e^{-\lambda} \cdot \lambda^k}{k!}$$

for this problem:

$$\lambda = (50 \times 0.03)$$

$$P(X=5) \approx \frac{e^{-\lambda} \cdot \lambda^5}{5!}$$

$$\approx \frac{e^{-1.5} \times (1.5)^5}{120}$$

$$P(X=5) \approx 0.184$$

Q19. Solution :-

Function :- A function f from a set A to a set B is a rule or

mapping that assigns to each element x in A exactly one element y in B , denoted as $f(x) = y$.

Types of Functions :-

1. Linear Function: A function $f(x)$ is linear if its graph is a straight line.

$$y_x \div f(x) = 2x+3$$

2. Quadratic Function :- A function $f(x)$ is quadratic if it is of the form $ax^2 + bx + c$, where $a \neq 0$.

Ex :- $f(x) = x^2 + 4x + 4$

3. Polynomial Function :- A function $f(x)$ is polynomial if it is a sum of terms, each of the form ax^n , where a is a constant and n is a non-negative integer.

Ex :- $f(x) = 3x^2 - 2x^2 + 5$

4. Exponential Function :- A function $f(x)$ is exponential if it has the form $f(x) = a \cdot b^x$ where a and b are constants.

Ex :- $f(x) = 2 \cdot 3^x$

5. Logarithmic Function :- A function $f(x)$ is logarithmic if it has the form $f(x) = \log_b(x)$ where b is the base of the logarithm.

Ex :- $f(x) = \log_2(x)$

6. Trigonometric Function :- Function like $\sin(x)$, $\cos(x)$, and examples of Trigonometric function.

Ex :- $f(x) = \sin(x)$

7. Piecewise Function :- A function defined by different expressions on different intervals of its domain.

Ex :- $f(x) = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$

Solution :- PMF of a Poisson distribution.

$$P(X=K) = \frac{e^{-\lambda} \cdot \lambda^K}{K!}$$

Where,

$P(X=K)$ [Prob. of observing K events]
 λ is the average rate of events per interval

e is the base natural log (≈ 2.71828)

$K!$ is the factorial of K

Given, $\lambda = 5$

Let, $Y = \text{sum of two independent poisson - distributed}$

$$Y = X_1 + X_2$$

$$P(Y=K) \Rightarrow \sum_{j=0}^K P(X_1=j) \cdot P(X_2=K-j)$$

for, $K < 2$, $K \Rightarrow 0, 1$

$$P(Y < 2) = P(Y=0) + P(Y=1)$$

$$\Rightarrow P(X_1=0) \cdot P(X_2=0) + P(X_1=1) \cdot P(X_2=0)$$

$$P(Y < 2) \Rightarrow \frac{e^{-5} \cdot 5^0}{0!} \times \frac{e^{-5} \cdot 5^0}{0!} + \frac{e^{-5} \cdot 5^1}{1!} \times \frac{e^{-5} \cdot 5^0}{0!}$$

$$P(Y < 2) \Rightarrow e^{-10} + 5e^{-10}$$

$$= e^{-10} (1+5)$$

$$P(Y < 2) \Rightarrow 6e^{-10} \approx 0$$

Q21. Solution :-

$$\mu \Rightarrow 72$$

$$\sigma \Rightarrow 15$$

a). $X = 60$

formula for Z-score

$$Z = \frac{X - \mu}{\sigma}$$

$$Z_a = \frac{60-72}{15}$$

$$\Rightarrow Z_a \Rightarrow -0.8$$

b) $X = 93$

$$Z_b = \frac{93-72}{\sigma}$$

$$\Rightarrow \frac{93-72}{15}$$

$$\Rightarrow Z_b \Rightarrow 1.4$$

c). $X = 72$

$$Z_c = \frac{72-72}{\sigma}$$

$$\Rightarrow \frac{72-72}{15}$$

$$\Rightarrow Z_c = 0$$

Q22.

Solution :-

Given,

$$\begin{aligned}\text{Mean } (\mu) &= 60 \\ \text{Standard Deviation } (\sigma) &= 5\end{aligned}$$

The Z-Score formula :-

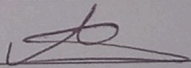
$$Z = \frac{X - \mu}{\sigma}$$

In this case, for $X = 60$

$$Z \Rightarrow \frac{60 - 60}{5}$$

$$Z = 0$$

Since the standard normal distribution is symmetric, the area to the right of $Z = 0$ is 0.5 (or 50%),

Therefore, 50% of students scored more than 60 marks. 

* Completed *